Barret Schloerke

Senior Software Engineer, Posit
`code[c("architect", "manager", "philosopher")]`
`data[c("wrangler", "explorer", "visualizer")]`

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Summary

Senior software engineer for Shiny, Posit's web framework for R and Python (1M+ monthly downloads). My statistician's skepticism (PhD) shapes how I build and test tools that turn data scientists into full-stack app developers. Highlights: built OpenTelemetry tracing into Shiny for R and Python, so users can see where their apps spend time; created shinyreact, so Shiny apps can use React-based UI; and cut Shiny's release testing from 3 weeks for 10 people to under 2 days for 2.

Experience

Senior Software Engineer, Posit · Remote — Jan 2022 – Present

- Core developer of Shiny for R and Python, Posit's flagship framework that brings users to its commercial products.
- Cut CI time 98%, measured by 4+ hours of testing finishing in a 4.5-minute run, by architecting Playwright controllers for every Shiny for Python component and spreading 1000+ tests over 170 jobs (5 Pythons × 3 browsers).
- Brought Shiny for R's best UI to Python by porting bslib's components: sidebars, cards, value boxes, tooltips, and more.
- Packaged shinylive for R and Python: run Shiny apps in the browser with no server, even in Quarto sites and slides.
- Enabled every Shiny user to trace apps with OpenTelemetry, build LLM-powered apps (querychat, shinychat), and pair agentic coding with React-based UI (shinyreact).

Software Engineer, RStudio (now Posit) · Remote — Jan 2018 – Dec 2021

- Cut release testing from 3 weeks across 10 people to under 2 days for 2 by architecting shinycoreci: nightly testing of 150+ apps on Mac, Windows, and Linux × 5 R versions. Built shiny-workflows, shared CI for every Shiny team package.
- Migrated Shiny's JavaScript to TypeScript, so every PR's client code is type-checked across all files (before, none was).
- Sustained plumber, reactlog, shinytest2, and dozens of other Shiny packages by owning releases, docs, and triage.

Tool Development Team, Gates Foundation: HBGDki · Remote — Apr 2016 – Dec 2018

- Produced custom summary statistics for 75+ datasets, feeding visualization apps that consultants used in work to prevent avoidable infant deaths.
- Protected sensitive study data by coordinating secure data-handling practices with five other data scientists.
- Standardized the team's R package development by creating and maintaining template R packages.

Researcher, DARPA XDATA and D3M grants, Purdue University · West Lafayette, IN — Sep 2016 – Dec 2017

- Delivered a web interface for subject matter experts to explore machine learning pipelines by partnering remotely with Hafen Consulting, Kitware, and KnowledgeVis.
- Led service integration across 15 partner working groups by serving as the lead point of contact.

Software Engineer, Metamarkets · San Francisco, CA — Jan 2011 – Jul 2012

- Built interactive D3 visualizations for analytics dashboards, backed by Druid, on DVL, a reactive data visualization library, in CoffeeScript.
- Wrote Node.js front-end servers that automated deploys, clusters, and company / user management, allowing anyone on the team to ship or debug the client experience.
- Reduced site startup time and server load by trimming data queries and startup work.

Data Scientist Intern, Novartis Pharmaceutical R&D · Basel, Switzerland — May 2008 – Aug 2008

- Equipped management with a live view of employee availability by building a Gantt chart tool.
- Standardized common graphics for the R&D team by writing a web tutorial that packaged them as reusable plotting layers.

Education

PhD, Statistics, Purdue University · West Lafayette, IN 2013 – 2017
- Advisors: William Cleveland and Ryan Hafen. Emphasis: large data visualization.
- National Science Foundation Graduate Research Fellow (2012 – 2017).

MS, Mathematical Statistics, Purdue University · West Lafayette, IN 2013 – 2014

PhD program, Statistics, Rice University · Houston, TX 2012
- Advisor: Hadley Wickham.

BS, Computer Engineering, Iowa State University · Ames, IA 2006 – 2010
- Statistics graphics research with Dianne Cook, Hadley Wickham, and Heike Hofmann.

Open source software

2,500+ merged pull requests across 170+ GitHub repositories, and 1,400+ reviews of others' pull requests, since 2018. Packages I maintain or author, by merged PRs:

`shiny` Python · author · 390+ PRs · A web development framework for Python.
- Agent Skills (2026): Skills shipped in the package so coding agents know how to write Shiny apps.
- Test mode (2026): test_server() and exported test values for fast tests without a browser.
- Session lifecycle (2026): session.destroy(), a soft teardown when a session closes, and session.allow_reconnect().
- Release train skill (2026): An Agent Skill that walks a py-shiny release from prep through PyPI, conda-forge, and shinylive.
- OpenTelemetry (2026): Traces and logs for sessions, reactivity, and outputs, so users can see where their apps spend time.
- Bookmarking (2025): Save and restore an app's state in the URL or on the server.
- Data frames (2024): render.data_frame with editing, selection, sorting, filtering, and styles.
- Output renderers (2024): shiny.render.renderer, a base class for writing your own output renderers.
- API reference (2023 – 2026): The quartodoc function reference and its runnable examples.
- UI components from bslib (2023 – 2024): Ported sidebars, cards, accordions, value boxes, tooltips, and popovers from R's bslib, so Python users get the same layout tools as R.
- Playwright testing (2023 – 2024): shiny.playwright controllers for every UI component, and the shiny.pytest plugin; 1000+ tests over 170 CI jobs finish in 4.5 minutes instead of 4+ hours.

`shinyreact` R, Python, TypeScript · maintainer · 170+ PRs · Client-Side 'React' Interface for 'shiny'
- TypeScript: npm package (2026): A dual build published as @posit-dev/shinyreact, usable with no server-side JS.
- TypeScript: Wire protocol (2026): One versioned protocol, checked by a handshake, shared with the R and Python packages.
- Python: Agent Skills (2026): Skills for building and converting apps with coding agents, shipped in the package.
- R: Agent Skills (2026): Skills for building and converting apps with coding agents, shipped in the package.
- R: Initial development (2026): Ported the Python API to R, then CRAN checks, a pkgdown site, and webR builds for 0.1.0; 2 other CRAN packages already use it.
- Python: Renderer dependencies (2026): Find and serve the HTML dependencies of Shiny outputs used inside React.
- Python: ui.tsx apps (2026): Write the whole page in React and serve it from Python.
- R: ui.tsx apps (2026): Write the whole page in React and serve it from R.
- TypeScript: React hooks (2026): useShinyInput / useShinyOutput hooks, and ShinyOutput for classic Shiny outputs.
- Python: Testing (2026): Playwright e2e tests, WireTap for wire payloads, and in-memory test_server().
- Python: Initial development (2026): Began as shinyjson, a JSON spec renderer, then turned React-first and dropped json-render.

`shiny` R, TypeScript · author · 160+ PRs · Web Application Framework for R
- R: MCP Apps (2026): Serve Shiny apps as MCP Apps, with author tools for LLM clients, so apps show up inside the chat in clients like Claude Desktop.
- R: Session lifecycle (2026): session$destroy(), a soft teardown when a session closes, and clean state on reconnect.
- R: OpenTelemetry (2025 – 2026): Traces and logs for sessions, reactivity, and outputs through {otel}, so users can see where their apps spend time.
- R: Quosures (2021): exprToFunction() and installExprFunction() accept quosures, including inject()ed ones.
- TypeScript: TypeScript (2021 – 2022): Ported shiny.js to TypeScript, moved the build to esbuild, and published the types; every PR's client code is now type-checked.
- R: devmode() (2020 – 2021): One switch that turns on the options you want while developing an app.
- R: testServer() (2020 – 2026): Test server functions and modules in memory against a mock session; no browser needed, so tests run much faster.
- R: Plot brushing (2018): Plot click and brush events carry CSS and image coordinates, and brushing works in WebKit.
- R: reactlog integration (2018 – 2020): Record every reactive event so reactlog can draw the reactive graph, letting users see the logic behind their app while debugging.

`GGally` R · maintainer · 150+ PRs · Extension to 'ggplot2'
- S7 (2025): Moved ggmatrix to S7 for ggplot2 4.0.
- Discrete stats (2020): Count, cross, table, and statistic plots for GGally 2.0.
- ggduo, ggnostic, and friends (2016 – 2017): Plot matrices for two groups of variables, model diagnostics, and time series.
- ggpairs and ggmatrix (2014 – 2017): The generalized pairs plot and the plot matrix behind it; the paper describing it has 120+ citations.

`shinytest2` R · maintainer · 150+ PRs · Testing for Shiny Applications
- Migration from shinytest (2022): migrate_from_shinytest() converts existing shinytest tests; shinytest was deprecated in May 2024 in favor of shinytest2.
- Screenshots (2021 – 2023): Consistent screenshots, compared with a kernel density smoother to ignore stray pixels from Chrome rounding and small font weight changes, so screenshot tests stop flaking.
- Test recorder (2021 – 2022): record_test() writes a test while you click around the app.
- AppDriver (2021 – 2022): Drive a Shiny app in headless Chrome: set inputs, click, and snapshot values.

`plumber` R · maintainer · 130+ PRs · An API Generator for R
- pr_*() API (2020): The pipe-friendly router API and consistent R6 methods of plumber 1.0.
- Docker (2020 – 2021): Docker images for running plumber APIs.
- Async (2019 – 2020): Promise and future support for long-running endpoints.
- Serializers and parsers (2019 – 2020): Body parsers chosen by content type, and serializers for many output formats.
- OpenAPI docs (2018 – 2020): OpenAPI v3 specs and pluggable docs pages (Swagger, RapiDoc, Redoc).

`reactlog` R · maintainer · 70+ PRs · Reactivity Visualizer for 'shiny'

`learnr` R · author · 70+ PRs · Interactive Tutorials for R
- Exercises (2019 – 2020): Safer exercise evaluation, checker functions, and clearer errors. Instructors use learnr for course materials, 29 CRAN packages use it, and its lessons shaped Posit Academy.
- Quiz questions (2019 – 2020): Questions rebuilt as Shiny modules, with a custom question API.

`promises` R · maintainer · 50+ PRs · Abstractions for Promise-Based Asynchronous Programming

`leaflet` R · author · 40+ PRs · Create Interactive Web Maps with the JavaScript 'Leaflet' Library

`shinyloadtest` R · maintainer · 30+ PRs · Load Test Shiny Applications

`htmltools` R · author · 30+ PRs · Tools for HTML

`shinylive` R · maintainer · 20+ PRs · Run 'shiny' Applications in the Browser

`shinyswatch` Python · author · 20+ PRs · Bootswatch + Bootstrap 5 themes for Shiny.

`chromote` R · author · 17 PRs · Headless Chrome Web Browser Interface

`sortable` R · author · 12 PRs · Drag-and-Drop in 'shiny' Apps with 'SortableJS'

`trelliscopejs` R · author · 11 PRs · Create Interactive Trelliscope Displays

`ellmer` R · author · 11 PRs · Chat with Large Language Models

`lmmpar` R · author · 9 PRs · Parallel Linear Mixed Model

`swagger` R · author · 9 PRs · Dynamically Generates Documentation from a 'Swagger' Compliant API

`leaflet.providers` R · maintainer · 8 PRs · Leaflet Providers

`autocogs` R · author · 8 PRs · Automatic Cognostic Summaries

`shinychat` R, Python · author · 8 PRs · Chat UI Component for 'shiny'

`lazyrmd` R · author · 7 PRs · Render R Markdown Outputs Lazily

`leaflet.extras` R · author · 7 PRs · Extra Functionality for 'leaflet' Package

`gqlr` R · maintainer · 7 PRs · 'GraphQL' Server in R

`packagedocs` R · author · 6 PRs · Build Website of Package Documentation

`sass` R · author · 6 PRs · Syntactically Awesome Style Sheets ('Sass')

`querychat` R, Python · author · 5 PRs · Filter and Query Data Frames in 'shiny' Using an LLM Chat Interface

`leaflet.esri` R · author · 4 PRs · 'ESRI' Bindings for the 'leaflet' Package

`gt` R · author · 3 PRs · Easily Create Presentation-Ready Display Tables

Also contributed to: bslib, shinylive, htmltools, posit-sdk, shinytest, httpuv, meetupr, rmarkdown, chatlas, websocket, flexdashboard, webshot2, cachem, later, vetiver, testthat, trelliscope, webshot, webdriver, fastmap, narwhals, plumber2, graphql, ggplot2, redoc, shinyvalidate, jquerylib, shinywidgets, routr, brand.yml.

Earlier research software

`rbokeh`, Purdue University · R interface to the Bokeh visualization library. 2016

DVL, Metamarkets · Functionally reactive JavaScript library binding data to visualizations. 2011 – 2012

`helpr`, Rice University · Friendlier HTML documentation for R, with search across packages. 2010

`cranvas`, Iowa State University · Interactive plotting canvas for R, drawing a million points in under a second. 2010

`tourr`, Iowa State University · Geodesic interpolation and basis generation for tour methods in R. 2009

`DescribeDisplay`, Iowa State University · Publication-quality R graphics from GGobi plots. 2009

`geozoo`, Iowa State University · Library of high-dimensional geometric objects, including the new Hyper Ring Torus. 2007

Publications

Gökalp Yavuz, F., and **B. Schloerke** (2020). Parallel computing in linear mixed models. *Computational Statistics 35(3): 1273–1289*.

Schloerke, B., H. Wickham, D. Cook, and H. Hofmann (2016). Escape from Boxland: Generating a Library of High-Dimensional Geometric Shapes. *The R Journal 8(2): 243–257*.

Emerson, J., W. Green, **B. Schloerke**, J. Crowley, D. Cook, H. Hofmann, and H. Wickham (2013). The Generalized Pairs Plot. *Journal of Computational and Graphical Statistics 22(1)*.

Hofmann, H., D. Cook, C. Kielion, **B. Schloerke**, J. Hobbs, A. Loy, L. Mosley, D. Rockoff, Y. Huang, D. Wrolstad, and T. Yin (2011). Delayed, Canceled, on Time, Boarding... Flying in the USA. *Journal of Computational and Graphical Statistics 20(2)*.

Mosley, L., D. Cook, H. Hofmann, C. Kielion, and **B. Schloerke** (2010). Monitoring the 2008 Election Visually. *Chance 23(3)*.

Talks & workshops

Building Custom Shiny UI with React · Johnson & Johnson Shiny Day (remote) Oct 2026

Beyond Bootstrap: Building Custom Shiny UI with React · posit::conf(2026) Sep 2026

OpenTelemetry in R: Capturing Traces, Logs, and Metrics for Shiny Applications · Open Source in Pharma Apr 2026

{GGally}: Success and Long-Haul Maintenance Strategies · ggplot2 extenders · video Nov 2025

Observability at scale: Monitoring Shiny Applications with OpenTelemetry · posit::conf(2025) · video Sep 2025

{plumber2}: Streamlining Web API Development in R · useR! 2025 · video Aug 2025

{querychat}: Natural language interfaces for interactive data graphics in Shiny · JSM 2025 Aug 2025

AI Kung-Fu : Training your LLM client in two function calls · ShinyConf 2025 Apr 2025

{shinylive}: Serverless Shiny applications workshop. An exercise in deploying your app to GitHub Pages · R/Pharma 2024 · video Oct 2024

Data frames in Shiny for Python · posit::conf(2024) · video — Aug 2024
{shinylive}: Serverless Shiny apps · ShinyConf 2024 · video — Apr 2024
Lessons learned from testing 2500+ Shiny Apps every day · ShinyConf 2023 · video — Mar 2023
{shinytest2}: Regression testing Shiny applications · Johnson & Johnson Shiny Day — Oct 2022
{shinytest2}: Regression testing Shiny applications · rstudio::conf(2022) · video — Jul 2022
{shinytest2}: Regression Testing for Shiny Applications · R/Pharma Shiny for FDA Submissions · video — May 2022
shinytest2: Regression Testing for Shiny Applications · ShinyConf 2022 · video — Apr 2022
plumber + future: Async Web APIs · Harvard R User Group · video — Aug 2021
An intro to plumber · Harvard R User Group · video — Aug 2021
plumber + future: Async Web APIs · rstudio::global(2021) · video — Jan 2021
Expanding R Horizons: Integrating R with Plumber APIs · RStudio Webinar · video — Oct 2020
Episode 14: Shining a Light on learnr · Shiny Developer Series · video — Oct 2020
Episode 13: Inside Plumber 1.0 · Shiny Developer Series · video — Sep 2020
Episode 12: reactlog · Shiny Developer Series · video — Sep 2020
Reactlog 2.0: Debugging the State of Shiny · Statistical Programming DC — Apr 2020
Reproducible Shiny apps with shinymeta · ABACUS 2019 — Nov 2019
Debug Large Data Apps: shiny + reactlog + sparklyr · Advanced R workshop, Northeastern University — May 2019
Reactlog 2.0: Debugging the State of Shiny · rstudio::conf(2019) · video — Jan 2019
Shiny · NASA Datanauts — Aug 2018
Cognostics: Metrics enabling detailed interactive visualization of big data · CSESC 2017 — Apr 2017
Web Scraping with R — Mar 2017
tidyverse[c("magrittr", "dplyr", "tidyr")] · Purdue Graduate Statistics Seminar — Jan 2017
Cognostics: Metrics enabling detailed interactive visualization of big data · JSM 2016 — Aug 2016
ggduo: Pairs plot for two group data · useR! 2016 — Jun 2016
Analysis and Visualization of Large Complex Data with Tessera · Spring Research Conference, IIT — Jun 2016
Analysis and Visualization of Large Complex Data with Tessera · Purdue Graduate Statistics Seminar — Apr 2016
Analysis and Visualization of Large Complex Data with Tessera · Iowa State Graphics Research Group — Mar 2016
Analysis and Visualization of Large Complex Data with Tessera · NUMBAT Seminar, Monash University — Mar 2016
gqlr = GraphQL + R: a data query language and runtime · WOMBAT 2016 — Feb 2016
Web Scraping with R · American Credit Acceptance — Oct 2015
Trelliscope: D&R visualization tool · Fields Institute Big Data Visualization · video — Feb 2015
ggplot2: displaying spatial and temporal data · Purdue Spatial Statistics Seminar — Oct 2014
git · Purdue Working Group — Jun 2014
Reducing Working Environment Inefficiencies · Purdue Graduate Statistics Seminar — Feb 2014
helpr: Help for R · Iowa State Working Group — Sep 2010
GGally: A Plot Matrix for All Variable Types · JSM 2010 — Aug 2010

Videos

Git Worktree-First Workflow + Conductor — Aug 2026
Shiny 1.2 Release: Wrap your data frame with Narwhals! — Oct 2024
Getting Started with {shinytest2} Part I · Example + basics — Jul 2022
Getting Started with {shinytest2} Part 3 · Using shiny.testmode in {shinytest2} — Jul 2022
Getting Started with {shinytest2} Part 2 · Exporting values — Jul 2022
Programming Games with Shiny · Roll the Dice: with Quosures! — Jun 2022
Programming Games with Shiny · Roll the Dice — May 2022
Programming Games with Shiny · Guess the Number — May 2022

Programming Games with Shiny · Dragon Realm — Mar 2022
Maximize computing resources using future_promise() — Feb 2022
{reactlog} Rundown — Feb 2022

Blog posts

All Together Now: Concurrent Async in Shiny for Python — Oct 2026
Shiny for Python 1.9 — Oct 2026
Genomes, Prompts, and Shiny: What I Built During My Summer at Posit — Oct 2026
Introducing shinyreact: React UI backed by a Shiny server — Sep 2026
Shiny for Python 1.8 — Sep 2026
Shiny updates: R 1.14, Python 1.7, bslib 0.12 — Aug 2026
Bringing OpenTelemetry to R in production — May 2026
Shiny for Python 1.6 brings toolbars and OpenTelemetry — Apr 2026
OpenTelemetry + Shiny for R v1.12 — Dec 2025
Tool Calling UI in shinychat — Nov 2025
promises v1.5.0 — Nov 2025
Shiny at posit::conf(2025) — Sep 2025
Posit at SciPy 2025 — Jul 2025
Shiny 1.4 brings bookmarking and Generative AI docs — Apr 2025
Shiny for Python 1.2.0 — Oct 2024
Testing Shiny Apps in Python — Oct 2024
Announcing Shiny for Python 1.0 — Jul 2024
Editable data tables in Shiny for Python 0.9.0 — May 2024
Introducing: Shiny Express — Jan 2024
Shiny for Python 0.6.1 — Dec 2023
Shiny for Python 0.6.0 — Nov 2023
Shiny for Python 0.5.0 — Aug 2023
Shiny for Python 0.4.0 — Jun 2023
Shiny for Python out of alpha — Apr 2023
plumber 1.1.0 — Mar 2021
Shiny 1.6 — Feb 2021
learnr 0.10.0 — Dec 2019
leaflet 2.0.0 — May 2018

Honors & awards

National Science Foundation Graduate Research Fellowship · the only Computational Statistics awardee of 13 statistics recipients in 2012 — 2012 – 2017
Visiting Scientific Researcher, Monash University · large data exploration tools with NUMBAT — 2016
Google Summer of Code, GGally::ggduo · pairs plots for two groups of variables — 2016
Best Dynamic / Interactive Visualization, Data Insight SF — 2011
NSF Graduate Research Fellowship, honorable mention — 2011
ASA Data Expo, 2nd place (Iowa State working group) — 2009

Teaching & mentoring

Internship Supervisor, Posit · Samuel Bharti: a bioinformatics Shiny showcase, 10 Shiny apps built on 5 packages — 2026
Workshop Teaching Assistant, rstudio::conf / posit::conf · Shiny in Production (2019), Applied Machine Learning (2020), Level Up with Shiny for R (2024), Shiny for R (2025) — 2019 – 2025

Internship Advisor, RStudio · Nischal Shrestha: gradethis, a companion package for grading learnr exercises — 2020

Internship Advisor, RStudio · JooYoung Seo: accessibility within Shiny for R — 2020

Internship Supervisor, RStudio · Leslie Huang: leaflet.providers and education materials — 2019

Internship Advisor, RStudio · Daniel Chen: gradethis, a companion package for grading learnr exercises — 2019

Instructor, R for Data Science Seminar, Purdue University — 2017

Teaching Assistant, Big Data Analytics: Statistics and Data Visualization, Purdue University — 2017

Teaching Assistant, Hadley Wickham's R Master Class, San Francisco and Chicago — 2011 – 2015

Service

President, Statistics Graduate Student Office, Purdue University · Ran a town hall, a graduate student mentor program, and an esteemed speaker event with Arthur Dempster. — 2016 – 2017

Graduate Student Mentor, Purdue University Statistics — 2014 – 2016

StatCom Assistant, Purdue University — 2014 – 2016

Skills

Languages	R, Python, TypeScript, JavaScript, HTML, CSS, SQL, Bash
Frameworks	Shiny (R and Python), React, plumber, htmltools, ggplot2, tidyverse
Testing & CI	Playwright, pytest, testthat, shinytest2, GitHub Actions
Publishing	CRAN, PyPI, npm, conda-forge, pkgdown, Quarto, GitHub Pages